



MODEL CONTEXT PROTOCOL

Interface Specification

Capabilities provided to a registered agent

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This document is the interface contract of `clarity-mcp`. It states the tools, profiles, and controls the product provides. A registered agent may read and propose. It may not execute.



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1 Purpose

Hutch Clarity explains a charge from system evidence, applies a remedy only when a versioned rule and the decision policy allow it, and issues a signed Trust Receipt for every executed plan.

`clarity-mcp` is the governed interface through which a registered agent uses those capabilities. The server name is `clarity`. The title presented to the client is Hutch Clarity. The server instruction is fixed:

Explain charges from evidence. Every answer must cite the case evidence or a rule. You cannot move money: the strongest action available is proposing a remedy the decision already allows, which a person or the customer then confirms.

The agent receives tool contracts. It does not receive credentials, database access, or an execution path. Amounts, causes, and eligibility are produced by rule packs and the decision policy. The agent writes and reads language around those facts.

2 What the product provides

Through this interface a registered agent can:

- read the evidence timeline of a case, including source completeness and the snapshot hash;
- read the cause assessment, the outcome, the ruled-out causes, and the actions the decision allows;
- read the plain description, legal basis, and required evidence of a published rule;
- read the public view of a Trust Receipt after one has been issued;
- read the safeguards recorded for the subscriber of a case;
- read the network status supplied for the subscriber of a case;
- search the published rule catalogue and receive cited passages;
- request a handoff to a person, with a reason code;
- propose one action the decision already allows, and receive a pending plan;
- on a staff session, read the desk queue ordered by money at stake.

Confirmation, approval, execution, receipt signing, bulk remedy, and policy publication are provided by other interfaces of the product. They are not tools on this server. Section 10 lists that boundary.

3 Architecture

`clarity-mcp` is a separate deployable from the Clarity API. It speaks MCP over Streamable HTTP at `/mcp`. It holds no database. It is an OAuth 2.1 resource server only: it validates tokens and never issues them.

A downstream call, where one is required, uses a new token obtained by RFC 8693 token exchange. The inbound client token is never forwarded. If token exchange is not configured, the downstream call is refused.

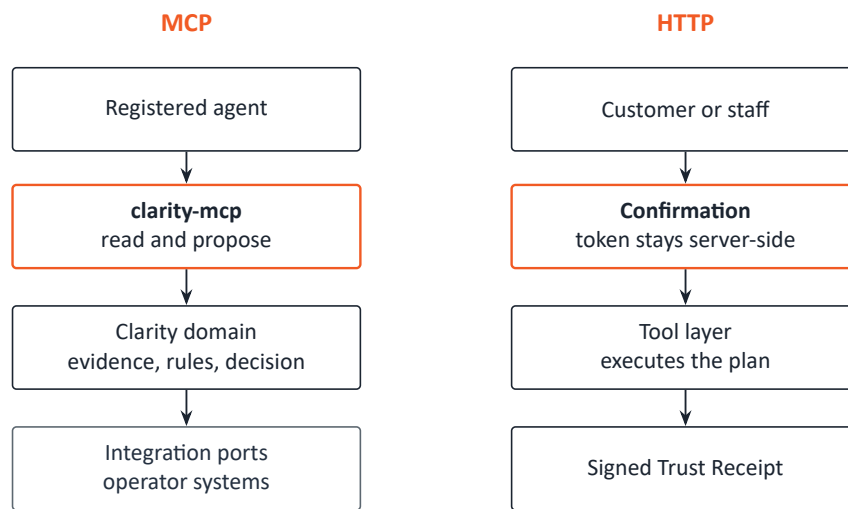


Figure 1: Two doors. An agent reaches Clarity only through `clarity-mcp`. Execution is reached only after confirmation on the HTTP interface. The two paths do not join inside the agent.

4 Transport and protocol

Item	Provided value
Protocol	Model Context Protocol, revision 2026-07-28
Transport	Streamable HTTP
Path	/mcp
Server name	clarity
Authentication	Bearer access token. A call with no token is refused (HTTP 401).
Resource indicator	RFC 8707. A token issued for another resource is refused.
Session state	Stateless. The server keeps no conversation memory.

Two MCP App resources are published for a host that can render them in a sandbox:

- `ui://clarity/why-card`, the explanation card for a case;
- `ui://clarity/receipt`, the receipt card.

5 Authentication and profiles

The profile is taken from the token scope. The client cannot select or widen it in the tool call. A token must carry exactly one profile scope. None is a denial. More than one is a denial.

Scope	Profile
<code>clarity.customer-assist</code>	<code>customer-assist</code> . Bound to one case.
<code>clarity.staff-assist</code>	<code>staff-assist</code> . Case and desk tools.
<code>clarity.analytics</code>	<code>analytics</code> . Rule explanation and knowledge search only.

A customer-assist token carries the case binding in the claim `clarity_case_id`. The binding is required. A customer session with no binding is refused on any call that names a case. A call for a different case is refused. The case identifier in the arguments is checked against the claim. It is not trusted on its own.



Staff and analytics sessions are not case-bound by that claim. Staff tools that take a `case_id` still pass the profile allowlist and the argument check.

Signature, issuer, audience, and expiry are verified before a tool runs. The same verifier serves the HTTP interface and this server, so the two doors accept the same class of token.

6 Safety levels

Level	On this interface	Meaning
L1	Provided	Read. No state change.
L2	Provided	Handoff. A person is requested. No money moves.
L3	Propose only	A pending plan for an action the decision already allows.
L4	Not provided	Bulk, policy, and regulator operations. Refused if reached.

An argument named `amount` or `amount_lkr` is refused with `AMOUNT_NOT_ACCEPTED`. Where a response includes an amount, that amount is copied from the decision or the plan. The caller did not supply it.

7 Tool catalogue

Ten tools are provided. The table is the allowlist. A name absent from it is `UNKNOWN_TOOL`.

Tool	Level	CA	SA	AN	Required arguments
<code>get_case_timeline</code>	L1	Yes	Yes		<code>case_id</code>
<code>get_cause_assessment</code>	L1	Yes	Yes		<code>case_id</code>
<code>explain_rule</code>	L1	Yes	Yes	Yes	<code>rule_id</code>
<code>get_trust_receipt</code>	L1	Yes	Yes		<code>case_id</code>
<code>get_customer_safeguards</code>	L1	Yes	Yes		<code>case_id</code>
<code>get_network_status</code>	L1	Yes	Yes		<code>case_id</code>
<code>search_knowledge</code>	L1	Yes	Yes	Yes	<code>query</code>
<code>request_handoff</code>	L2	Yes	Yes		<code>case_id</code> , <code>reason</code>
<code>propose_action</code>	L3	Yes	Yes		<code>case_id</code> , <code>action_type</code>
<code>get_desk_queue</code>	L1		Yes		none

CA is customer-assist, SA is staff-assist, AN is analytics. A blank cell means the tool is not in that profile.

7.1 `get_case_timeline`

Returns the evidence snapshot for the case.

- `case_id`
- `snapshot_hash`
- `events[]`: `event_id`, `source`, `event_type`, `occurred_at`, `amount_lkr` when the event carries one
- `sources`: completeness of each source, one of `complete`, `partial`, `missing`

If no snapshot is stored, the server builds one from the case record before answering. Customer text is not evidence. The events are system records.



7.2 `get_cause_assessment`

Returns the decision already recorded for the case. If the case has not been evaluated, the call is `CASE_NOT_READY`.

- `outcome`: `AUTO_FIX`, `ONE_TAP_FIX`, `STAFF_APPROVAL`, `EXPLAIN_ONLY`, or `HANDOFF`
- `amount_lkr`, from the decision record
- `cause`: `rule_id`, `rule_version`, `confidence`, when a top cause exists
- `ruled_out []`: rule identifiers the decision excluded
- `allowed_actions []`: the only action types a later proposal may name
- `rationale`

7.3 `explain_rule`

Returns the published description of one rule pack.

- `rule_id`, `version`, `description`
- `legal_basis`, when the pack states one
- `required_evidence []`
- `allowed_actions []`

An unknown identifier is `UNKNOWN_RULE`.

7.4 `get_trust_receipt`

Returns the public view of the receipt issued for the case. If no receipt has been issued, the call is `CASE_NOT_READY`. The agent does not create, sign, or alter a receipt. A receipt is emitted when a plan completes, on the execution path in Figure 1.

7.5 `get_customer_safeguards`

Returns the subscriber mask and the safeguard actions recorded on the case execution:

- `BLOCK_MERCHANT_UNTIL_OPTIN`
- `ENABLE_DATA_STOP`
- `SET_SPEND_CAP`
- `ENABLE_FUP_ALERTS`

The response lists types present on the execution. It does not change them.

7.6 `get_network_status`

Returns the coverage and outage state held for the subscriber of the case, together with `case_id`. The value is supplied by the network integration port. In this release that port is a labelled simulation. The field must not be read as a measurement from a live radio network.



7.7 search_knowledge

Searches the published rule catalogue. Draft rules are excluded. The query is masked before it is recorded.

- Input: query
- Output: query after masking, corpus = rule-catalogue, and up to five chunks
- Each chunk carries text, source_id of the form rule_id@version, version, legal_basis when stated, and score

A passage is citable because source_id names a governed rule pack. A legal basis is copied from the pack. It is never inferred.

7.8 request_handoff

Records a request to route the case to a person.

- Input: case_id, reason (stored up to 200 characters)
- Output: case_id, status = handoff_requested, reason, requested_by

7.9 propose_action

Creates a pending plan for one action type. This is the strongest operation on the interface. It does not execute the plan, and it accepts no amount.

- Input: case_id, action_type
- Output: plan_id, case_id, status, summary, amount_lkr taken from the plan, and a note that the plan is pending confirmation

action_type must be one of the following. Any other value is UNKNOWN_ACTION. The decision must already list the type in allowed_actions. A proposal the decision does not allow is denied by the tool layer and audited.

Action type	Level	Effect, after confirmation
REFUND	L3	Return the decided amount.
DEACTIVATE_VAS	L3	End the value-added subscription.
BLOCK_MERCHANT_UNTIL_OPTIN	L3	Block the merchant until the customer opts in.
SET_SPEND_CAP	L2	Set the spend cap the decision allows.
ENABLE_DATA_STOP	L2	Stop further data spend.
ENABLE_FUP_ALERTS	L2	Enable fair-use alerts.
CREATE_SUPPORT_CASE	L2	Open a support case.
SEND_NOTIFICATION	L2	Send an approved template.

The plan remains pending until the customer confirms it, or a staff member approves it, on the HTTP interface. That step mints and spends a single-use confirmation token. The token is never returned to the agent.

7.10 get_desk_queue

Staff profile only. Returns cases that have a decision and no execution, ordered by money at stake, descending. Each row carries case_id, case_no, outcome, and money_at_stake_lkr.



1. **Read.** The agent calls `get_case_timeline` and `get_cause_assessment`.
2. **Propose.** `propose_action` names an `action_type` from `allowed_actions`. The response is a pending `plan_id`.
3. **Confirm, outside MCP.** The customer or a staff member confirms on the HTTP interface. The server mints the confirmation token and spends it.
4. **Execute.** The tool layer runs the plan. A repeated confirmation returns the same outcome and the same receipt.
5. **Prove.** A signed Trust Receipt is issued. The agent may then call `get_trust_receipt`.

Figure 2: The agent stops at a pending plan. Money moves only after confirmation, on a path the agent cannot call.

8 Proposal, confirmation, and execution

9 Denials

Every call is audited, including a denial. The audit stores the tool name, safety level, principal, profile, case identifier, a hash of the arguments, the decision (allowed, denied, or error), and the error code. Argument values are hashed. They are not retained as customer text.

Code	When it is returned
UNKNOWN_TOOL	The name is not in the catalogue.
NOT_IN_PROFILE	The token's profile does not include the tool.
INVALID_ARGUMENTS	A required argument is missing.
AMOUNT_NOT_ACCEPTED	The caller sent an amount.
SESSION_NOT_BOUND	A customer session has no case claim.
NOT_AUTHORISED_FOR_CASE	The case is not the one in the token.
LEVEL_NOT_EXPOSED	The operation is L4.
UNKNOWN_RULE	<code>explain_rule</code> was given an unknown rule.
UNKNOWN_ACTION	<code>action_type</code> is not in the action vocabulary.
CASE_NOT_FOUND	The case identifier does not exist.
CASE_NOT_READY	The case has no decision, or no receipt, for a tool that requires one.
NO_PROFILE_SCOPE	The token has no profile scope.
AMBIGUOUS_PROFILE	The token has more than one profile scope.

A refusal the server expected is returned to the client as a tool error. An unauthenticated call ends at HTTP 401 and receives no tool result.

10 Capabilities provided elsewhere

The following are part of Hutch Clarity and are intentionally absent from `clarity-mcp`.



Capability	Where it is provided
Confirm a plan	HTTP interface. Mints and spends the confirmation token.
Staff approval and step-up	HTTP interface, under the staff role.
Execute a refund or a service change	Tool layer, after a valid confirmation.
Sign and chain a Trust Receipt	Receipt service, on completion of a plan.
Auto-fix without a person	Deterministic path, only where the decision is <code>AUTO_FIX</code> . Never from an agent.
Bulk remedy, policy publish, regulator export	Staff operations. Classed L4. Never over MCP.
Open a case and attach evidence	HTTP interface and the integration ports.

An integration that needs a registered agent to explain a charge, cite a rule, hand the case to a person, or propose an allowed remedy uses this specification. An integration that needs to confirm, approve, or execute uses the HTTP interface. Operator systems of record are reached through the integration ports, on the domain side of Figure 1.

11 Document control

Version	Date	Status	Note
1.0	5 October 2026	Issued	First issue of the MCP interface specification. Catalogue matches the ten tools provided by <code>clarity-mcp</code> .